

Exacting embeddings of countable structures

Virtually exacting cardinals, large countable ordinals,
and the exact role of the Kunen inconsistency

Research note accompanying the synthesis docs/research-synthesis; prepared for Vladimir Reshetnikov,
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Summary

Question. Is there a connection between large cardinals (LC) and large countable ordinals (LCO) that supports theorems and not only analogies, for the cardinals studied in this repository?

Answer proposed here. Yes, through *virtual* large cardinals in the sense of Gitman and Schindler [3]: embeddings that exist only in a forcing extension, equivalently embeddings between countable structures. We define *virtually exacting* cardinals and the corresponding *L-exacting* countable ordinals and prove:

1. (Theorem 3.1) a Kunen-free core: the critical points κ_n of any virtual witness are inaccessible, and κ_0 is virtually rank-into-rank. The proofs are those already machine-checked in the Lean development, which never use that the embedding belongs to the universe.
2. (Theorem 4.1) If $0^\#$ exists then in L the cardinal i_ω is virtually exacting with a cofinal critical sequence and is *inaccessible*; every limit Silver indiscernible is virtually exacting. So the three basic facts about exacting cardinals — cofinality ω , singularity, $V \neq \text{HOD}$ — all fail virtually, and virtual exactingness is consistent with $V = L$.
3. (Theorem 5.1) What survives when the critical sequence is cofinal: λ is a strong limit, fixed small sets lie below the critical point, and λ is regular in HOD_{V_λ} . Hence a dichotomy: either $\text{cf}(\lambda) = \omega$ and $V \neq \text{HOD}_{V_\lambda}$ (the situation of a genuine exacting cardinal), or λ is a regular limit of inaccessibles (the situation in L). The Kunen inconsistency is used for exactly one thing: to put the critical sequence into V .
4. (Theorem 6.1) The least L -exacting countable ordinal λ_{ex} , if it exists, is a limit of ordinals κ with $L_\kappa \models \text{ZFC}$ and lies below the least stable ordinal; it exists if $0^\#$ does. This gives a new family of large countable ordinals, one for each virtual large cardinal notion.
5. (Proposition 5.3) By contrast *ultra*exactingness does not virtualize: a virtual ultraexacting witness yields a genuine rank-into-rank embedding in V .

Status. Conventional proofs, not refereed. Items 1, 3 and 5, and Propositions 7.2 and 7.1, are formalized in Lean over an arbitrary transitive set model (Section 8), including the two clauses — Theorems 3.1(4) and 5.1(3) — that the first version of this note left out; items 2 and 4 are not formalized, because they quantify over L . I searched for “virtually exacting” and found nothing, but exacting cardinals date from 2024 and I cannot exclude unpublished work. The results are modest: they are what one obtains by running the arguments of the synthesis through the Gitman–Schindler machinery, and their main value is to locate precisely where the Kunen inconsistency enters.

1 Analogies between large cardinals and large countable ordinals

The following table collects the connections; the last four rows are the ones this repository contributes to. “S” refers to the synthesis.

Large cardinals	Large countable ordinals	Nature of the link
inaccessible, Mahlo, weakly compact, ...	Ω, I, M, K in collapsing functions ψ	scaffolding: each cardinal can be replaced by its recursive analogue
regular / Mahlo / Π_n^1 -indescribable	admissible / recursively Mahlo / Π_{n+1} -reflecting	the same reflection scheme, second order over V_κ versus first order over L_α
$V_\alpha < V_\beta, C^{(n)}$ cardinals	$L_\alpha <_{\Sigma_1} L_\beta$, stable ordinals	elementary substructures of the hierarchy
consistency strength of T	proof-theoretic ordinal $ T $	calibration of Π_1^1 consequences; known only far below ZFC
measurable, 0^+ , Woodin	mice, $0^\#$, heights and ranks of countable iterable structures	inner model theory; determinacy
λ regular in HOD_{V_λ} , rigidity (S §8, §12)	admissibility, Σ_1^1 -boundedness at ω_1^{CK}	same theorem shape: a definable family of short cofinal sets is empty or as large as possible (S Prop. 15.1)
Prikry core at an ultraextending λ (S Thm. 13.3)	$\langle \kappa_n \rangle$ generic over the ω th iterate (Bukovský–Dehornoy)	the same theorem with a single embedding in place of an iteration
$Q_\lambda = D_\lambda / \equiv^*$ and definable choice (S §9, §12, §14)	$\mathcal{P}(\omega)/\text{fin}, E_0$, Baire measurability	different answers: CF versus a global fixed point (S Prop. 15.3)
extending witnesses $j : X \rightarrow V_\zeta$	embeddings of countable L_ζ ; Silver indiscernibles	<i>this note</i> : the virtual version lives in L and on countable ordinals
orbit classes of $j \restriction \lambda$ (S Lemma 12.1)	fundamental sequences from iterating a normal function	Proposition 7.2

2 Definitions

Recall (Aguilera–Bagaria–Lücke [1]) that λ is *extending* if for every $\zeta > \lambda$ there are $X < V_\zeta$ with $V_\lambda \cup \{\lambda\} \subseteq X$ and an elementary $j : X \rightarrow V_\zeta$ with $j(\lambda) = \lambda$ and $j \restriction \lambda \neq \text{id}$. Such a λ has cofinality ω , is a strong limit, is regular in HOD_{V_λ} , and therefore $V \neq \text{HOD}$.

Definition 2.1. A *virtual extending witness* at λ of height $\zeta > \lambda$ is a pair (X, j) in a set-forcing extension $V[G]$ such that $X < V_\zeta^V$, $V_\lambda^V \cup \{\lambda\} \subseteq X$, and $j : X \rightarrow V_\zeta^V$ is elementary with $j(\lambda) = \lambda$ and $j \restriction \lambda \neq \text{id}$. Put $\kappa = \text{crit}(j)$, $\kappa_0 = \kappa$, $\kappa_{n+1} = j(\kappa_n)$ (each $\kappa_n < \lambda$ belongs to X) and $\kappa_\omega = \sup_n \kappa_n \leq \lambda$. The witness is of *cofinal type* if $\kappa_\omega = \lambda$ and of *bounded type* otherwise. The cardinal λ is *virtually extending* if it has virtual witnesses of every height $\zeta > \lambda$, and *virtually extending of cofinal type* if it has virtual witnesses of cofinal type of unboundedly many limit heights.

A genuine extending witness is a virtual one with G trivial; and, by the Kunen inconsistency, it is of cofinal type (S Lemma 3.1). The structures V_ζ^V and the requirement $V_\lambda^V \subseteq X$ refer to V , as in all of [3]; only X and j are new.

Definition 2.2. A countable ordinal λ is *L-extending* if there are a countable $\zeta > \lambda$ with $L_\zeta \models \text{ZFC}$, a set $X < L_\zeta$ with $V_\lambda^{L_\zeta} \cup \{\lambda\} \subseteq X$, and an elementary $j : X \rightarrow L_\zeta$ with $j(\lambda) = \lambda$ and $j \restriction \lambda \neq \text{id}$. Cofinal and bounded type are defined as before.

By condensation X is isomorphic to some $L_{\bar{\zeta}}$, so an L -exacting witness is the same as a pair of elementary embeddings $i, k : L_{\bar{\zeta}} \rightarrow L_{\zeta}$ with $i \upharpoonright V_{\lambda}^{L_{\bar{\zeta}}} = \text{id}$, $i(\lambda) = k(\lambda) = \lambda$ and $\text{crit}(k) < \lambda$: one embedding that does not see V_{λ} and one that acts on it.

Lemma 2.3 (Gitman–Schindler [3, Lemma 3.1]). Let $W \subseteq V$ be transitive models of ZFC and $M, N \in W$ first-order structures with M countable in W . If V has an elementary embedding $M \rightarrow N$, so does W , and it can be required to agree with the given one on finitely many points.

The proof (a tree of finite partial elementary maps is ill-founded in V , hence in W) works for structures in a countable language; we use it with constants for the elements of V_{λ} .

Proposition 2.4. Let $\lambda < \zeta$ be countable ordinals with $L_{\zeta} \models \text{ZFC}$ and $V_{\zeta}^L = L_{\zeta}$ (for instance ζ a strong limit cardinal of L). Then λ is L -exacting with a witness of height ζ if and only if $L \models \text{“}\lambda \text{ has a virtual exacting witness of height } \zeta\text{”}$; and the types agree.

Proof. Let G be $\text{Col}(\omega, L_{\zeta})$ -generic over V . If (i, k) is a witness in V , apply Lemma 2.3 to $L[G] \subseteq V[G]$ and the structures $(L_{\bar{\zeta}}, \in, x)_{x \in V_{\lambda} \cup \{\lambda\}}$, $(L_{\zeta}, \in, x)_{x \in V_{\lambda} \cup \{\lambda\}}$ for i , and $(L_{\bar{\zeta}}, \in, \lambda, \kappa)$, $(L_{\zeta}, \in, \lambda, k(\kappa))$ for k : both embeddings exist in the generic extension $L[G]$ of L . Conversely a witness in a generic extension of L is a witness in $V[G]$, and Lemma 2.3 for $V \subseteq V[G]$ returns one in V . Cofinality of the critical sequence can be arranged by prescribing $k(\kappa_n) = \kappa_{n+1}$ level by level in the tree. $\square$

3 The Kunen-free core

Theorem 3.1. Let (X, j) be a virtual exacting witness at λ of limit height ζ , with critical sequence $\langle \kappa_n \rangle$.

1. $j(n) = n$ for $n < \omega$, $j(\omega) = \omega$, and $\kappa > \omega$.
2. $j(x) = x$ for every $x \subseteq \mu < \kappa$ in V .
3. For $\mu < \kappa$ there is in V no surjection of $\mathcal{P}(\mu)$ onto κ and no cofinal map $\mu \rightarrow \kappa$. So κ is inaccessible in V , and so is every κ_n .
4. $e = j \upharpoonright V_{\lambda}$ is an elementary embedding $V_{\lambda} \rightarrow V_{\lambda}$ in $V[G]$; hence κ is virtually rank-into-rank [3, Def. 2.4], and therefore an ω -iterable limit of ω -iterable cardinals and a virtually n -huge* limit of such cardinals for every n [3, Thm. 4.15, 4.20].
5. If $S \in X$, $S \subseteq \lambda$, $j(S) = S$ and $|S|^V < \kappa$, then $j^{\omega} S = S$ and S consists of fixed points of j .

The same holds for L -exacting witnesses, with L_{ζ} in place of V .

Proof. (1)–(3) and (5) are the statements `jOrd_nat`, `jOrd_omega`, `omega_le_crit`, `jv_subset_fixed`, `no_surj_pow_crit`, `slSem_critSeq`, `crit_regular` and the first half of `fixed_small_subset` of the Lean development, whose proofs use the embedding only through the two schemes “ $V_{\zeta} \models \varphi[\vec{x}] \iff V_{\zeta} \models \varphi[j(\vec{x})]$ for $\vec{x} \in X$ ” and “existential statements with parameters in X have witnesses in X ”; the axiom audit confirms that they do not depend on the (admitted) Kunen inconsistency. For instance, for (3): a surjection $f : \mathcal{P}(\mu) \rightarrow \kappa$ can be found in X ; then $j(f)$ maps $\mathcal{P}(\mu)$ onto $j(\kappa) \ni \kappa$, while $j(f)(x) = j(f(x)) = f(x) < \kappa$ for every $x \subseteq \mu$ by (2) — a contradiction. For regularity, a cofinal $f : \mu \rightarrow \kappa$ in X has $j(f) = f$ pointwise, but $j(f)$ is cofinal in $j(\kappa) > \kappa$. (4): $j(V_{\lambda}) = V_{\lambda}$ because $j(\lambda) = \lambda$, and $V_{\lambda} \models \varphi[x]$ iff $V_{\zeta} \models \text{“}V_{\lambda} \models \varphi[x]\text{”}$. $\square$

4 What fails: virtually exacting cardinals in L

Theorem 4.1. Assume $0^\#$ exists and let $\langle i_\alpha \rangle$ enumerate the Silver indiscernibles.

1. $L \models "i_\omega \text{ is virtually exacting of cofinal type}"$. In L , i_ω is inaccessible (indeed ineffable and more).
2. $L \models "i_\gamma \text{ is virtually exacting}"$ for every limit ordinal γ ; for $\gamma > \omega$ there are witnesses of bounded type, and witnesses for which $j(\kappa_\omega) \neq \kappa_\omega$.
3. Consequently $\text{ZFC} + V = L + \text{"there is a virtually exacting cardinal of cofinal type"}$ is consistent relative to $0^\#$, and none of the statements " $\text{cf}(\lambda) = \omega$ ", " λ is singular", " $V \neq \text{HOD}$ " follows from virtual exactingness.

Proof. (1) Let $\lambda = i_\omega$ and let $\zeta > \lambda$ be arbitrary. Write $\zeta = t^L(i_{a_1}, \dots, i_{a_p}, i_{b_1}, \dots, i_{b_q})$ with $a_1 < \dots < a_p < \omega \leq b_1 < \dots$, and let $m > a_p$. Let H be the Skolem hull in L of $\{i_n : n < \omega\} \cup \{i_\omega\} \cup \{i_{b_1}, \dots, i_{b_q}\} \cup \{i_\beta : \beta \in B\}$ for a fixed infinite set B of indices above all b_l . Every $x \in L_{i_\omega}$ is of the form $t^L(i_{n_1}, \dots, i_{n_r}, i_{\beta_1}, \dots, i_{\beta_s})$ with β 's in B , because the value of such a term, when it lies below i_ω , does not depend on the indiscernibles above i_ω ; so $L_\lambda \cup \{\lambda, \zeta\} \subseteq H$, and $V_\lambda^L = L_\lambda$. The map σ that sends $i_n \mapsto i_{n+1}$ for $n \geq m$ and fixes the other generators is order preserving, so it induces an elementary $j : H \rightarrow H$ with $j(\lambda) = \lambda$, $j(\zeta) = \zeta$, $\text{crit}(j) = i_m$ and critical sequence $i_m, i_{m+1}, \dots$, cofinal in λ . Put $X = H \cap V_\zeta^L$. Since V_ζ^L is definable in $H < L$ from ζ , $X < V_\zeta^L$ and $j \upharpoonright X : X \rightarrow X$ is elementary for V_ζ^L . This is a witness in V for the structure V_ζ^L ; by Lemma 2.3, applied as in the proof of Proposition 2.4 in $V^{\text{Col}(\omega, V_\zeta^L)}$ and its submodel $L[G]$, there is one in a generic extension of L . (2) For $\lambda = i_\gamma$ use all i_α , $\alpha < \gamma$, as generators, the shift $i_n \mapsto i_{n+1}$ above m on the first ω of them and the identity elsewhere: then $\kappa_\omega = i_\omega < \lambda$ and $j(\kappa_\omega) = \kappa_\omega$. For $\gamma = \omega \cdot 2$ the map $i_\alpha \mapsto i_{\alpha+1}$ ($m \leq \alpha < \omega \cdot 2$) gives $\kappa_\omega = i_\omega$ and $j(i_\omega) = i_{\omega+1}$. (3) follows from (1) and [1, Thm. 2.10], since $L \models V = \text{HOD}$. $\square$

So for virtual embeddings the supremum of the critical sequence, the fixed cardinal λ , and its cofinality come apart completely, as Gitman and Schindler observed for rank-into-rank embeddings. What Theorem 4.1 adds is that *even when the critical sequence is cofinal in λ* the cardinal can be regular: the sequence $\langle i_n \rangle$ is simply not in L .

5 What survives: cofinal type

Theorem 5.1. Let λ be virtually exacting of cofinal type.

1. λ is a strong limit cardinal and a limit of inaccessible virtually rank-into-rank cardinals.
2. If (X, j) is a witness of cofinal type, $S \in X$, $S \subseteq \lambda$, $j(S) = S$ and $|S|^V < \lambda$, then $S \subseteq \text{crit}(j)$; and there is no nonempty $\mathcal{F} \in X$ with $j(\mathcal{F}) = \mathcal{F}$, $|\mathcal{F}|^V < \lambda$, consisting of cofinal subsets of λ of order type below λ .
3. λ is regular in HOD_{V_λ} ; more precisely every nonempty $\mathcal{A} \in \text{OD}_{V_\lambda}$ of cofinal subsets of λ of order type below λ has $|\mathcal{A}| \geq \lambda$.
4. Hence exactly one of the following holds: λ is singular in V and $V \neq \text{HOD}_{V_\lambda}$; or λ is inaccessible. If $V = \text{HOD}_{V_\lambda}$, in particular if $V = L$, the second alternative holds.

Proof. (1) By Theorem 3.1(3) each κ_n is a strong limit, and $\lambda = \sup_n \kappa_n$ for the witnesses of cofinal type. (2) Every $\alpha \in [\kappa, \lambda)$ is moved by j : if $\kappa_n \leq \alpha < \kappa_{n+1}$ then $j(\alpha) \geq \kappa_{n+1}$. Now $\mu = |S|^V$ is definable from S in V_ζ (ζ is a limit above λ), so $\mu \in X$ and $j(\mu) = \mu$, whence $\mu < \kappa$; by Theorem 3.1(5) S consists of

fixed points, so $S \subseteq \kappa$. For a family $\mathcal{F}$, let τ be the least order type of a member and u the union of the members of order type τ : $u \in X$ is fixed, cofinal, and $|u| \leq |\mathcal{F}| \cdot |\tau| < \lambda$, a contradiction. These are the proofs of S Lemma 8.4; the first is fixed_small_subset in Lean, with the cofinality of the critical sequence now a hypothesis instead of a consequence of Kunen's theorem. (3) Call a nonempty $\mathcal{A} \in \text{OD}_{V_\lambda}$ of short cofinal subsets of λ with $|\mathcal{A}| < \lambda$ a *counterexample*. Every member of OD_{V_λ} is definable over some V_θ from ordinals below θ and one parameter in V_λ . Suppose counterexamples exist and let θ_0 be least such that one of them is so definable over V_{θ_0} . Take a witness (X, j) of cofinal type of limit height $\zeta > \theta_0 + \omega$. Satisfaction over set structures is absolute for V_ζ , so V_ζ defines θ_0 from λ ; thus $\theta_0 \in X$ and $j(\theta_0) = \theta_0$. In the same way the following are definable from λ in V_ζ , hence fixed: the least cardinality ν_0 of a counterexample definable over V_{θ_0} ; the least rank ρ_0 of a parameter defining one of cardinality ν_0 ; and the least pair $(\varphi, \vec{\alpha})$, in a canonical well-order of formulas and finite sequences of ordinals, that does so with some parameter of rank ρ_0 . Being fixed ordinals below λ , ν_0 and ρ_0 are below κ . Let $\mathcal{F}$ be the union of all counterexamples of cardinality ν_0 defined over V_{θ_0} by $(\varphi, \vec{\alpha})$ from a parameter of rank ρ_0 . Then $\mathcal{F} \in X$ is nonempty and fixed, consists of short cofinal sets, and $|\mathcal{F}| \leq |V_{\rho_0+1}| \cdot \nu_0 < \lambda$ by (1). This contradicts (2). (4) If λ is singular in V , then V has a short cofinal subset of λ and HOD_{V_λ} has none by (3), so $V \neq \text{HOD}_{V_\lambda}$; if λ is regular it is inaccessible by (1). $\square$

Corollary 5.2. For a cardinal λ the following are equivalent: (a) λ is exacting; (b) λ has, for every ζ , a virtual exacting witness that belongs to V . In that case the witnesses are of cofinal type and $\text{cf}(\lambda) = \omega$. In L , by contrast, a virtually exacting cardinal of cofinal type is inaccessible and no witness belongs to L .

Thus, of the properties of an exacting cardinal, the Kunen inconsistency is responsible for *one*: that the critical sequence is a set of the universe and is cofinal. Everything else in S §3–§8 that is proved from a single witness — strong limit, the fixed-small-sets lemma, regularity in HOD_{V_λ} , the projection-width theorem — holds for virtual witnesses of cofinal type, in L if one likes.

Proposition 5.3. Suppose (X, j) is a virtual exacting witness at λ and $j \upharpoonright V_\lambda \in X$ (a virtual *ultraexacting* witness). Then $e = j \upharpoonright V_\lambda$ belongs to V , so V has a nontrivial elementary $e : V_\lambda \rightarrow V_\lambda$, the witness is of cofinal type, $\text{cf}(\lambda) = \omega$, and $V \neq L$.

Proof. $X \subseteq V_\zeta^V \subseteq V$. Now apply the Kunen inconsistency in V to e : if $\kappa_\omega < \lambda$ then $e \upharpoonright V_{\kappa_\omega+2}$ is a nontrivial elementary embedding of $V_{\kappa_\omega+2}$ into itself. $\square$

So exactingness has a virtual form compatible with L and ultraexactingness has none. This matches the consistency-strength picture of [2], where ultraexacting cardinals are tied to I0 while exacting cardinals are weaker, and it explains why all results of S §12–§13 that use $e \in X$ (orbit classes in X , the internal normal measure) have no counterpart in L other than the classical theory of indiscernibles.

6 Large countable ordinals

Theorem 6.1. Suppose some countable ordinal is L -exacting, and let λ_{ex} be the least one.

1. λ_{ex} is the same whether computed in V or in L , and it is countable in L .
2. $\lambda_{\text{ex}} < \sigma$, where σ is the least stable ordinal ($L_\sigma \prec_{\Sigma_1} L$); indeed a witness belongs to L_σ .
3. λ_{ex} is a limit of ordinals κ with $L_\kappa \models \text{ZFC}$; in particular it is far above ω_1^{CK} , the recursively Mahlo and Π_n -reflecting ordinals, the nonprojectible ordinals, and the least α with $L_\alpha \models \text{ZFC}$.
4. If $0^\#$ exists then L -exacting ordinals exist, of both types, and $\lambda_{\text{ex}} < \omega_1^L < i_0$.

Proof. (1) “Some countable ordinal is L -exacting” is a Σ_2^1 sentence (there is a real coding a well-founded model of $V = L$, with the embeddings), so it holds in L ; let λ_1 be least in L . For $\lambda < \lambda_1$, which is countable in L , “ λ is L -exacting” is Σ_2^1 in a real of L coding λ , hence fails in V ; and λ_1 is L -exacting in V . (2) In $L_{\omega_1^L}$ the existence of a witness is a Σ_1 statement, so it holds in L_σ . (3) Let (X, j) be a witness at λ_{ex} of height ζ and $\kappa = \text{crit}(j)$. If $\beta \in (\kappa, \lambda_{\text{ex}})$ were fixed by j , then (X, j) would be a witness at β , contradicting minimality; so no ordinal in $(\kappa, \lambda_{\text{ex}})$ is fixed. By Theorem 3.1(3) for L_ζ , κ is inaccessible in L_ζ , so the supremum β of the inaccessibles of L_ζ below λ_{ex} is at least $j(\kappa) > \kappa$; it is definable in L_ζ from λ_{ex} , hence fixed, hence equal to λ_{ex} . And $L_\kappa = V_\kappa^{L_\zeta} \models \text{ZFC}$ for every inaccessible κ of L_ζ . (4) Theorem 4.1 with $\zeta = i_{\omega,2}$, which is countable and satisfies $L_\zeta = V_\zeta^L < L$; then (1) and (2). $\square$

The argument is not specific to exacting cardinals. For every virtual large cardinal notion A of [3] that is witnessed by embeddings between set-sized structures, “the least countable α such that some countable $L_\zeta \models \text{ZFC}$ has embeddings witnessing that α is A ” is a countable ordinal between the least α with $L_\alpha \models \text{ZFC}$ and the least stable ordinal, it is the same in V and in L , and it exists if $0^\#$ does. The order of these ordinals reflects the virtual large cardinal hierarchy. They are large countable ordinals in the sense of the last section of the usual lists, defined by an embedding property rather than by reflection or stability; I have not found them discussed.

7 Critical sequences and fundamental sequences

A cofinal ω -sequence in a countable limit ordinal is what proof theory calls a fundamental sequence, and the classes of Q_λ are fundamental sequences up to finite change. The orbit structure of S Lemma 12.1 is ordinal arithmetic and holds for any normal function.

Proposition 7.1. Let e map $[\kappa, \lambda)$ into itself with $e(\xi) > \xi$. Then every $\xi \in [\kappa, \lambda)$ is $e^n(r)$ for some $n < \omega$ and some root r .

Proposition 7.2. Let $e : \lambda \rightarrow \lambda$ be strictly increasing with $e(\xi) > \xi$ for $\xi \in [\kappa, \lambda)$ and $\sup_n e^n(\kappa) = \lambda$. Then the forward orbits $a_r = \{e^n(r) : n < \omega\}$ of the roots $r \in [\kappa, \lambda) \setminus e''[\kappa, \lambda)$ partition $[\kappa, \lambda)$ into cofinal sets of order type ω ; and if $r \leq r' < e(r)$ then $e^n(r) \leq e^n(r') < e^{n+1}(r)$ for all n . So any two orbits are interleaved with bounded index displacement, and they represent pairwise different classes of Q_λ .

The interleaving is interleave in Cardinals/KunenFree.lean. For $e(\xi) = \omega^\xi$, $\kappa = 0$, $\lambda = \varepsilon_0$ the orbit of 0 is the standard fundamental sequence $0, 1, \omega, \omega^\omega, \dots$; for $e(\xi) = \varphi_\xi(0)$ one gets that of Γ_0 ; and for $e = j \upharpoonright \lambda$ the critical sequence. In each case the least root gives the canonical sequence and the other roots give “parallel” ones. In the large-cardinal case S Theorem 13.3(2) singles out the least root by a property of the class alone (its tail filter is normal, the others are not). Whether there is an analogous intrinsic characterization of the standard fundamental sequences of proof theory among their parallels — a notoriously informal matter — is a question this analogy suggests but does not answer.

8 What is formalized

The Lean development in Cardinals/Virtual/ verifies the parts of this note that are theorems of ZFC about a single witness. In Lean’s ZFSet every function on a set is itself a set, so a virtual embedding cannot be modelled by an embedding into a rank of the ambient universe — there the Kunen inconsistency applies. Instead the ambient universe plays the role of $V[G]$ and an arbitrary

transitive set A plays the role of the ground structure $(V_\zeta^V, \text{ or a countable } L_\zeta)$:

Twitness $A \lambda Y :$ $X < A, (A \cap V_\lambda) \cup \{\lambda\} \subseteq X, j : X \rightarrow A$ elementary, $j(\lambda) = \lambda, j \upharpoonright \lambda \neq \text{id}$,

with *no* requirement that $j \in A$. A genuine witness is the special case $A = V_\alpha$ ($\text{RelWitness.toTWitness}$), with the same critical point and critical sequence. Since A may be countable, all notions of size are taken internal to A : “small” means “the image of an ordinal below λ under a function belonging to A ”, and the power set is $\mathcal{P}^A(y) = \mathcal{P}(y) \cap A$.

Statement	Lean	Status
Definitions 2.1, 2.2	TWitness, CofinalType, toTWitness	definitions; genuine witnesses are the case $A = V_\alpha$
Theorem 3.1(1)	jOrd_nat, jOrd_omega, omega_le_crit	proved
Theorem 3.1(2)	jv_subset_fixed	proved
Theorem 3.1(3)	no_surj_pow_crit, slSem_critSeq, crit_regular	proved (inaccessibility of every κ_n in the sense of A)
Theorem 3.1(4)	relTo, relTo_spec, vA_fixed, fixed_below_crit, restriction_elementary, restriction_moves_crit	proved, with one further admitted input (below). The clause “hence κ is virtually rank-into-rank” is the Gitman–Schindler <i>definition</i> applied to the conclusion, and is not formalized
Theorem 3.1(5), 5.1(2)	least_surj_fixed, fixed_small_subset, no_small_fixed_family	proved from CofinalType as a <i>hypothesis</i>
Theorem 5.1(3)	no_short_cofinal_low_rank, no_small_family_low_rank, fixed_param, least_ordinal_fixed	proved for parameters of rank below λ (which includes every ordinal $< \lambda$) together with λ itself, using the same admitted input as Theorem 3.1(4) and nothing else. least_ordinal_fixed supplies the device for a parameter that is a <i>large</i> ordinal, and is itself free of admitted statements
Lemma 2.3	elementary_iff_assignment, exists_free_bound, assignment_iff_branch	the combinatorial core: an embedding of a countable structure is a branch of a tree of finite approximations. The absoluteness step itself is a metatheorem about two models and is not expressible in one Lean universe
Proposition 5.3	IsUltra, cofinalType_of_ultra	proved from an admitted in-model Kunen inconsistency (Published.kunen_in_model, referenced to Kunen 1971 and Kanamori Cor. 23.14)
Theorem 6.1(3)	restrictTo, no_fixed_point_of_minimal	proved: a fixed point of j above $\text{crit}(j)$ carries a witness, so at a least witnessed ordinal nothing above the critical point is fixed
Propositions 7.2, 7.1	interleave, exists_root	proved
Theorems 4.1, 6.1(1),(2),(4)	—	not formalized: they quantify over L , $0^\#$, Silver indiscernibles and stable ordinals, and neither Mathlib nor ProveIt has the constructible hierarchy

Two devices made the last two clauses possible, and neither needs a satisfaction predicate inside A — which is what the first version of this note reported as the obstruction.

Relativization. Elementarity of the restriction $j \upharpoonright V_\lambda^A$ is a statement about one formula at a time, so instead of a truth predicate it is enough to translate each formula: $\text{relTo } k \Vdash_E$ restricts every quantifier

of φ to the variable v_k , and `relTo_spec` proves that $A \models (\text{relTo } k \ \varphi)[\vec{x}]$ iff $(x_k, \epsilon) \models \varphi[\vec{x}]$. Combined with the fixedness of the stage V_λ^A this gives Theorem 3.1(4). The one admitted ingredient is that the cumulative hierarchy is definable in a transitive model of ZF, so that V_ρ^A is the unique solution of a formula with parameter ρ ; it is a textbook fact (Jech, Chapters 6 and 12).

Making parameters fixed. Theorem 5.1(3) needs the defining data of a counterexample to be fixed by j . Two mechanisms do this, and the Lean development has both. First, `fixed_below_crit`: j fixes *every* set of rank below `crit(j)` — proved by induction on the rank, using the fixed stages again, and false at `crit(j)` itself. So a parameter becomes fixed as soon as the witness is chosen after it, which is the hypothesis `HasHighWitnesses` (for genuine ultraexacting cardinals, [1, Lemma 3.2]; for virtual ones, part of the hypothesis, exactly as in this note). This is the minimal-rank-parameter device of report R18 in the synthesis, and it yields the theorem for all parameters of rank below λ . Secondly, `least_ordinal_fixed`: if φ , with parameters already fixed, is satisfied in A only by ordinals, then the *least* ordinal satisfying it is fixed — however large, and although j moves cofinally many ordinals. The formula `leastF` $\ddot{\text{I}}\ddot{\text{E}}$ expressing “ v_0 satisfies φ and no element of v_0 does” is built from φ at the meta-level, so again no truth predicate appears. That is the least-counterexample device of the proof of Theorem 5.1(3) in isolation. What is *not* carried out is the bookkeeping that iterates it over the finitely many ordinal parameters and the defining rank of an arbitrary OD_{V_λ} definition; that step is routine and adds nothing mathematical.

The axiom audit (`Cardinals/Audit.lean`) records exactly which admitted statements each theorem uses, and the picture is the following. Theorem 3.1(1)–(3), Theorem 5.1(2), Lemma 2.3, Propositions 7.1 and 7.2, and the two devices `relTo_spec` and `least_ordinal_fixed` depend only on `propext`, `Classical.choice` and `Quot.sound` — on no admitted statement, and in particular on no form of the Kunen inconsistency. That is the precise content of the claim that the core of Sections 3 and 5 is Kunen-free: it is not an inspection of the paper proofs but a property of the machine-checked ones. Theorem 3.1(4) and Theorem 5.1(3) depend, through `fixed_below_crit`, on the definability of the cumulative hierarchy, and on nothing else; `restriction_elementary` taken by itself, with the fixedness of V_λ^A as a hypothesis, does not. Proposition 5.3 depends on the admitted in-model Kunen inconsistency, as it must.

9 Questions

Question 9.1. (i) What is the consistency strength of a virtually exacting cardinal (of cofinal type)? It is at least that of a virtually rank-into-rank cardinal and at most that of $0^\#$; the argument of [3, Thm. 4.17] suggests an upper bound near an ω -Erdős cardinal, with the same difficulty they note (the fixed point may be the supremum of the indiscernibles). (ii) Can a virtually exacting cardinal of cofinal type be singular of uncountable cofinality? (iii) Is there a virtual version of cover exactingness for which the barrier S Theorem 5.1 holds, and is it consistent with a strongly compact cardinal below λ ? (iv) Where exactly does λ_{ex} sit relative to the ordinals attached to the other virtual large cardinals, and relative to the least α such that L_α has a nontrivial elementary self-embedding?

References

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- [3] V. Gitman, R. Schindler. Virtual large cardinals. *Ann. Pure Appl. Logic* **169** (2018), 1317–1334. Definition 2.4, Lemma 3.1, Theorems 4.15, 4.17, 4.20. (Numbering checked against the authors' version.)